Appendix: return models and baseline contrasts

Love Vaishnav | 2026-10-14

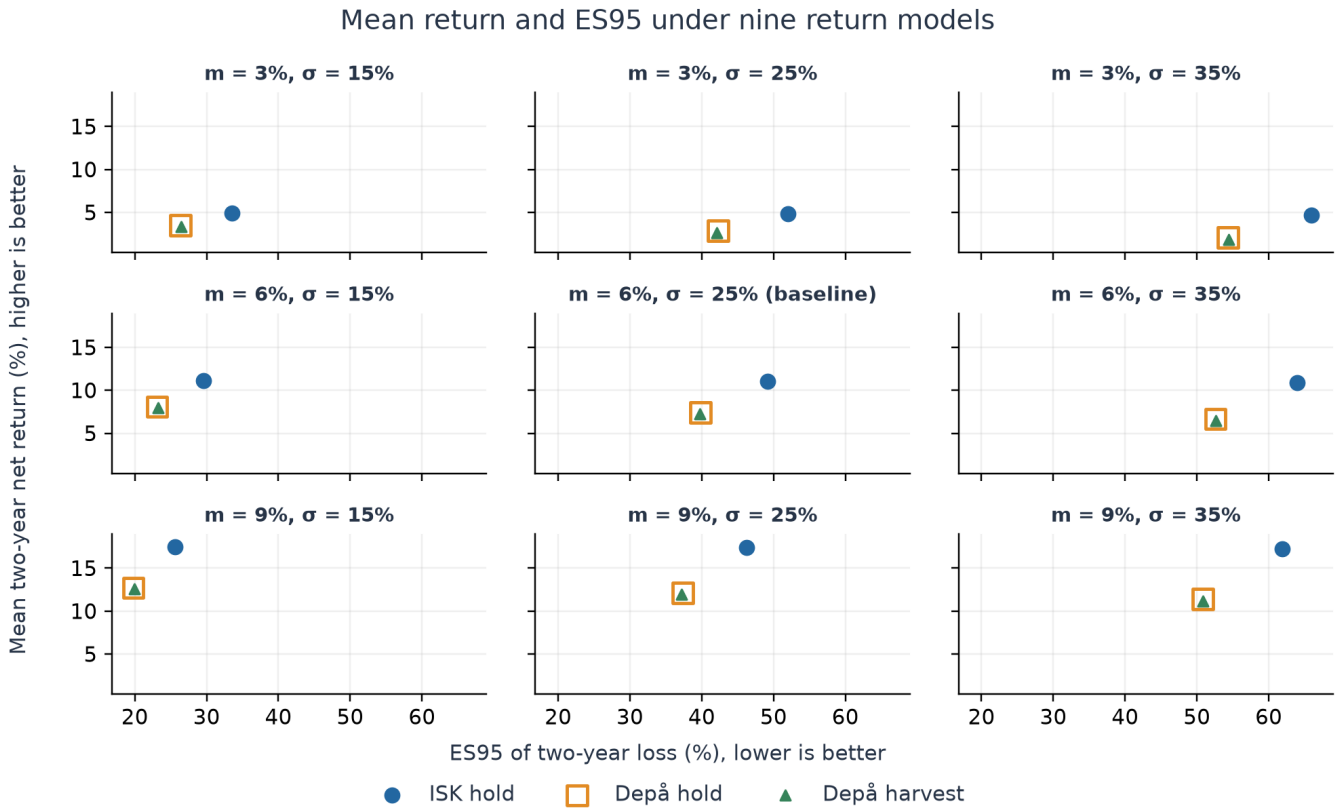


Figure 4. Mean return and ES95 for nine illustrative return models (m, σ), all other inputs at baseline, using the same 20,000 innovations. Hollow squares (depå hold) and small triangles (depå harvest) nearly coincide because their differences are a fraction of a percentage point; tables/contrasts.csv gives them with intervals.

Baseline contrast	Mean difference, pp [MC interval]	ES95 difference, pp [MC interval]
H − D	-0.194 [-0.201, -0.186]	-0.164 [-0.202, -0.131]
H − I	-3.765 [-3.913, -3.622]	-9.474 [-9.564, -9.386]
D − I	-3.572 [-3.720, -3.430]	-9.310 [-9.394, -9.227]

Expected shortfall. For an integrable loss L, ES95 = $\min_v \{v + E[(L - v)^+]/0.05\}$ [P4]. With N = 20,000 equally weighted paths the estimator averages exactly the worst 5% of the sample, i.e. the largest 1,000 losses; ties are kept, and a fractional boundary observation is used when 0.05N is not an integer. A difference of two ES95 values is not the ES95 of the paired differences.